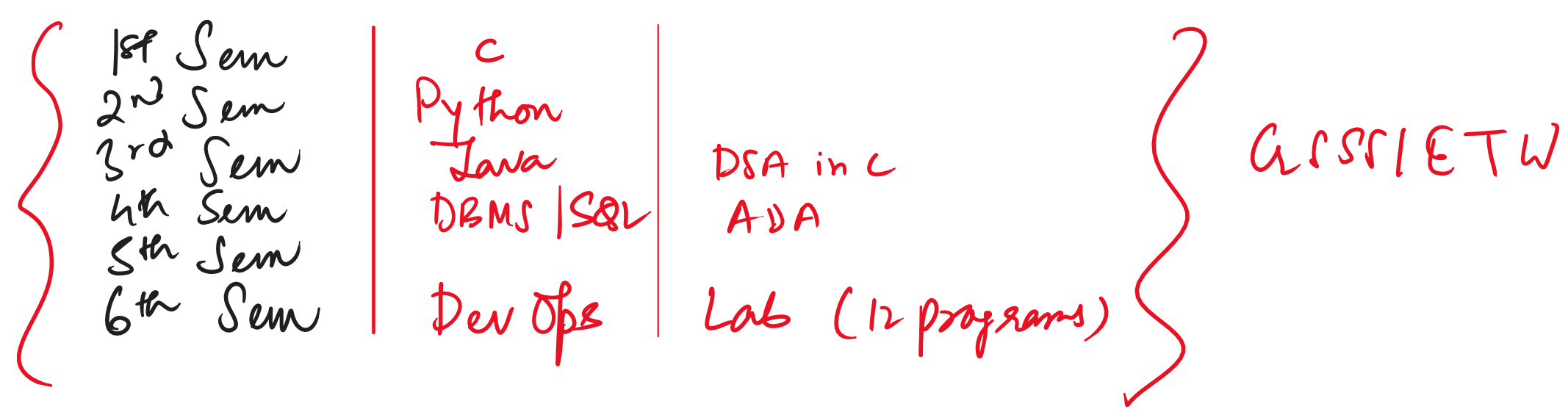




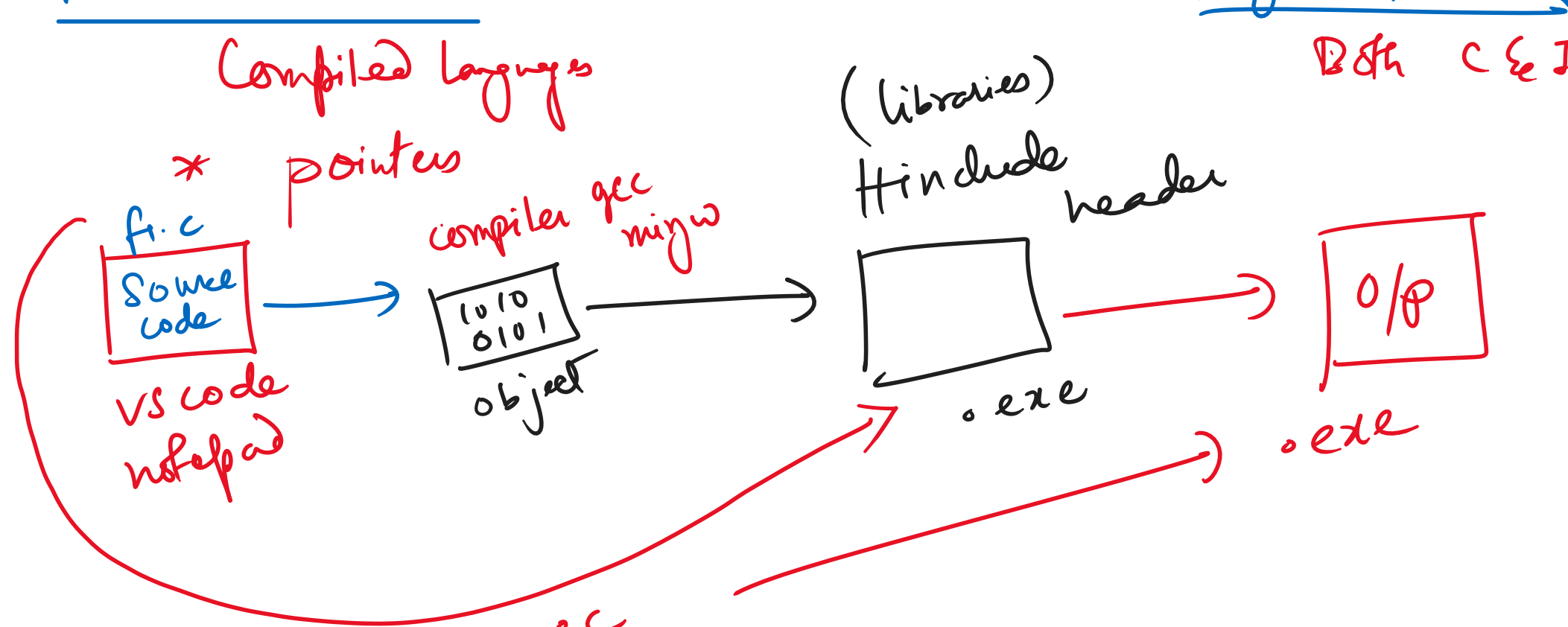
Placement Companies : $\rightarrow$ { Standard Chartered | Visa | Mercedes | Pure Storage | Infosys }

History : $C \rightarrow 1972$
 Dennis Ritchie (1940) $\rightarrow A + \text{mod}^n$
 $B + \text{mod}^n$
 C
 ANSI $\rightarrow C$

1st std Ver $\rightarrow 1989$
 $C + \text{SIMULA} = \boxed{C++}$ Bjarne Stroustrup
 $\downarrow$ RVST * y (WORA)
 $\uparrow$

| C & C++ |

| Python | Java |



{ Data Types of C } $\text{bool} \rightarrow \langle \text{stdbool.h} \rangle$

Primary	Void	Enumerated	Derived
Integer	void	enum (4B)	Array
Short $\rightarrow 2B \rightarrow (-128 \text{ to } 127)$		Named Constants	String
int $\rightarrow 4B \rightarrow (-2^{31} \text{ to } 2^{31}-1)$		Enumeration	Struct
long $\rightarrow 8B$		MON - SUN	Union
		JAN - DEC	Pointers
Floating		R, B, G, Y	
float 4B		H, L, M	
double 8B		integer - default	
Character			
1B			

(C++) $\rightarrow$ (OOP)

- * String
- * bool
- * auto
- * namespaces
- * no format specifiers
- * new & delete $\rightarrow$ DMA

#include <iostream>
 using namespace std;
 :: scope resolution operator

add(a, b) {
 Ankita
 Bangalore
 (App) {
 Same
 Codebase
 }
 f.c .cpp
 add(a, b) {
 Anusha
 Mumbai
 }

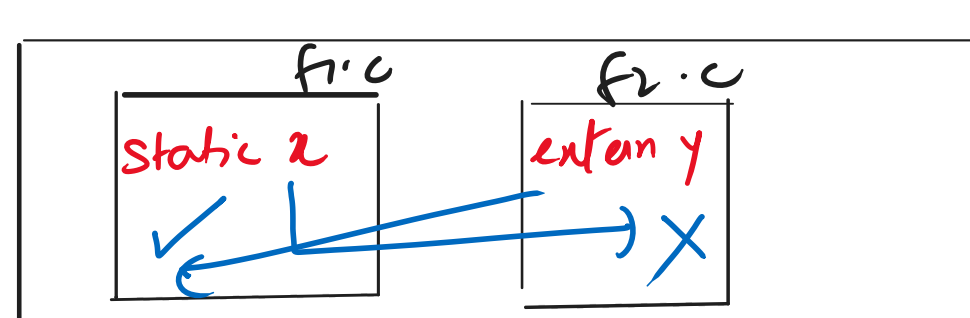
4 Storage Classes:

- (I) static
- (II) extern
- (III) register
- (IV) auto

scope
 * global
 * global RAM
 local

lifetime/validity
 Inside file (Multiple files)
 Inside Folder
 $\rightarrow$ No pointers allowed

Dangling Pointer



void a() {
 {
 }
 }
 void b() {
 {
 }
 }

1995 $\rightarrow$ James Gosling (C+C++)

(Java) $\rightarrow$ (JDBC) $\rightarrow$ API $\rightarrow$

(SHA-256) $\rightarrow$ (S12) $\rightarrow$ (JDK-25)

$\rightarrow$ OOP $\rightarrow$ Sun Micro Systems

Coffee (Java)

(Oracle)

(1991) Guido van Rossum

Monty {Python}
 Flying Circus

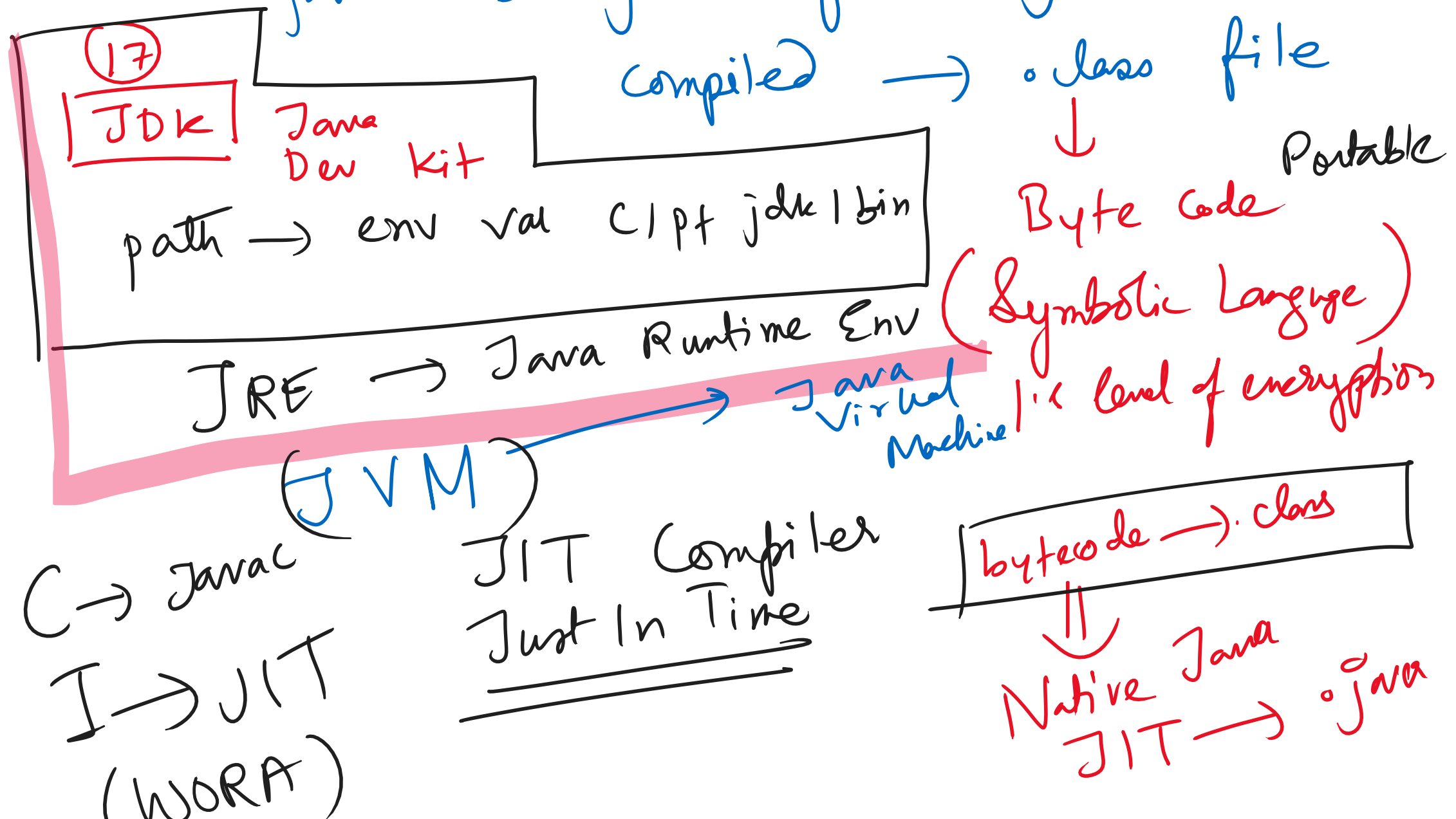
#include <stdio.h>
 int main() {
 printf("CISSEIETW");
 return 0;
 }

All class names $\rightarrow$ Pascal case
 underscore case $\rightarrow$ rate-of-interest

C&I Anatomy of a Pascal Case Java Program
 public class CISSEIETW {
 public static void main(String[] args) {
 System.out.println("CISSEIETW");
 }
 }
 Pascal Rate Of Interest $\rightarrow$ classes, interface or constants
 Camel rate of Interest $\rightarrow$ variables & methods

WORA C&I Run-code Platform Independent $\rightarrow$ W, M, L, U, A
 .java $\rightarrow$ java filename.java

OS $\rightarrow$ path of Java



C $\rightarrow$ javac
 I $\rightarrow$ JIT (WORA)
 JIT Compiler Just In Time

bytecode $\rightarrow$.class
 $\downarrow$
 Native Java
 JIT $\rightarrow$.java